Bioresource Technology 349 (2022) 126868

Contents lists available at ScienceDirect

Bioresource Technology

journal homepage: www.elsevier.com/locate/biortech

Signature of dissolved organic matter and microbial communities based on different oxygen levels response during distillers dried grains with solubles plus sugarcane pith co-fermentations

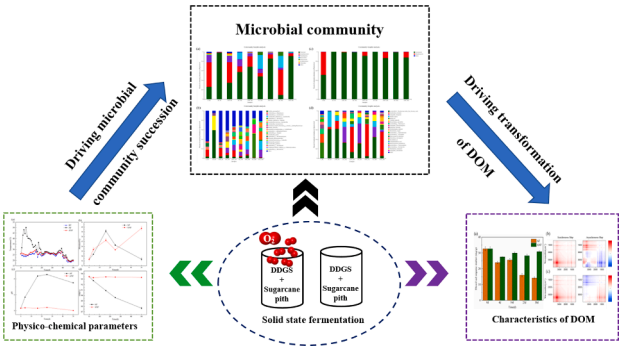
Yiwu Wang [a], Xiaolan Li [a], Kecheng Li [a], Yite Huang [a], Hongxiang Yang [a], Pengfei Zhu [a], Zhanyou Chi [b], Yongping Xu [b], Qunliang Li [a,*]

[a] School of Chemistry and Chemical Engineering, Guangxi University, Nanning 530004, China
[b] School of Bioengineering, Dalian University of Technology, Dalian 116024, China

HIGHLIGHTS

- Signature of DOM and associated microbial community were studied.
- Phenols and alcohols were consumed first in AF, while lipids and proteins first in ANF.
- *Firmicutes*, *Actinobacteriota* and *Ascomycota* dominated the fermentation process.
- pH, OM and DOM were the main factors driving microbial community succession.

GRAPHICAL ABSTRACT



ARTICLE INFO

Keywords:
DDGS
Sugarcane pith
Co-formentation
Microbial community
Dissolved organic matter

ABSTRACT

The objective of this study was to investigate the relationship between dissolved organic matter (DOM) and microbial communities during the co-fermentation of distillers dried grains with solubles (DDGS) and sugarcane pith at different oxygen levels. In aerobic fermentation (AF), the content of DOM decreased from 32.61 mg/g to 14.14 mg/g, and decreased from 32.61 mg/g to 30.83 mg/g in anaerobic fermentation (ANF). Phenols and alcohols were consumed first in AF, while lipids and proteins were consumed first in ANF. Degradation rates of cellulose, hemicellulose and lignin in AF (6.67%, 39.93%, 36.50%) were higher than those in ANF (0.69%, 18.36%, 9.12%). *Firmicutes*, *Actinobacteriota* and *Ascomycota* were the main phyla in community. Distance-based redundancy analysis showed that pH, organic matter (OM) and DOM were the main driving factors of microbial community succession.

* Corresponding author.
E-mail address: liqunliang231@gxu.edu.cn (Q. Li).

https://doi.org/10.1016/j.biortech.2022.126868
Received 15 January 2022; Received in revised form 14 February 2022; Accepted 15 February 2022
Available online 17 February 2022
0960-8524/© 2022 Elsevier Ltd. All rights reserved.

1. Introduction

Since the Industrial revolution, with the extensive use of coal, oil, natural gas and other fossil energy, the ecological environment of the earth has been destroyed. Greenhouse gas emissions pose a serious threat to the sustainable development of human society (Muhammad et al., 2014). In order to ensure national energy security, many countries have issued policies to encourage the development of fuel ethanol industry (Zabed et al., 2017). According to the American Renewable Fuels Association (RFA), China accounted for about 4.0% of the world's fuel ethanol production, making it the third largest ethanol producer (Wu et al., 2021). The production mode of bioethanol can be divided into dry grinding and wet grinding, and dry grinding is the main mode of production (Iram et al., 2021). Because of the high cost and water consumption, most ethanol plants in China usually use the improved dry grinding method to produce ethanol. The by-product called dry distiller's grains with solubles (DDGS) was obtained by dry grinding in the ethanol plant, and its rational use could improve the benefits of the corn ethanol refineries.

DDGS was widely used in animal husbandry, aquaculture, and poultry feed because it was rich in crude protein, fat, and vitamins and other nutrients (Iram et al., 2020). There are anti-nutritional factors such as restricted amino acids and phytic acid in DDGS, which affect the absorption and utilization of nutrients by monogastric animals (Sun et al., 2021a). Solid-state fermentation was used to improve the nutritional characteristics, digestibility and palatability of feed, and macromolecular substances and anti-nutritional factors were transformed into high-quality, non-toxic feed raw materials or commercial feed under the metabolism of natural or artificial microorganisms (Yang et al., 2021; Li et al., 2020a, Li et al., 2020b). At present, nitrogen-rich DDGS with lignocellulosic waste (rich in fermentable sugars), were co-fermented to produce nutrient-rich pig feed (Sun et al., 2021a). Guangxi sugar industry produces a large amount of sugarcane pith (rich in fermentable sugars) every year. Burning or discarding sugarcane pith directly is a waste of resources. Sugarcane pith can be used as the substrate of solid state fermentation to produce animal feed, which can increase the added value and protect the environment.

Dissolved organic matter (DOM) was a complex organic mixture composed of amino acids, sugars, polyphenols, humus and other compounds, which contains carboxyl, quinone, aldehyde and other functional groups, so it could participate in adsorption, electron transfer reaction, microbial metabolism and chelation (Wang et al., 2014; Nebbioso and Piccolo, 2013). Dissolved organic matter included dissolved organic carbon (DOC), dissolved organic nitrogen (DON) and dissolved organic phosphorus (DOP). Attention to the dynamic changes of DOM is helpful to further understand the role of DOM in carbon, nitrogen and phosphorus biochemical processes. Nowadays, two-dimensional correlation spectroscopy (2D-COS) played an important role in exploring the evolution and interaction mechanism of organic compounds at the molecular level. The combination of 2D-COS technology and Fourier Transform infrared spectroscopy (FTIR) has made an important contribution to the study of DOM, which is helpful to analyze the structure and changes of DOM (Niu et al., 2021; Chen et al., 2019). In addition to FTIR, 2D-COS technology can also be used to analyze DOM in combination with UV–visible spectrum, fluorescence spectrum, excitation-emission matrix-parallel factor (EEM-PARAFAC) and Raman spectrum (Chen et al., 2019). In the process of aerobic composting and anaerobic digestion, microorganisms could use DOM to grow and reproduce, and microbial metabolic pathways such as glycolysis, anaerobic carbon fixation, lignin decomposition and so on were also closely related to DOM (Li et al., 2018; Jiang et al., 2019; Suominen et al, 2021; Zhu et al., 2020). In other words, the properties of DOM affected the metabolic capacity of microbial communities, and the composition of dissolved organic matter was influenced by microbial metabolic reactions (Fasching et al., 2020). Under aerobic or anaerobic treatment, although the same raw materials are used, the system has a unique microbial community and biotransformation pathway, which affects the final product (Fernandez-Bayo et al., 2018). For example, *Aspergillus* and *Bacillus subtilis* secrete protease, xylanase, pectinase and so on under aerobic conditions, while lactic acid bacteria can produce organic acids under anaerobic conditions. The succession of microbial communities in the substrate was promoted or inhibited at different oxygen levels, which in turn affected the composition of DOM and ultimately the quality of solid fermented feed. Thus, it is necessary to compare and analyze the aerobic and anaerobic fermentation of DDGS and sugarcane pith. Studies on the relationship between DOM and microbial community were mainly focused on the environment and composting, while there were few studies on solid state fermentation of animal feed. Therefore, in order to provide more information, the relationship between environmental factors, DOM and microbial community during DDGS and sugarcane pith co-fermentation under different oxygen levels was explored.

The aim of this work was: (1) to use two-dimensional correlation spectroscopy (2D-COS) FTIR to study the change sequence of DOM characteristic functional groups during the co-fermentation of sugarcane pith and DDGS; (2) to use dynamic analysis of microbial community to obtained the dominant fungi and bacteria in different stages of fermentation;(3) to establish the relationship between physicochemical parameters, DOM and microbial community. This study provided a new idea for DDGS and sugarcane pith co-fermentation process and optimization of fermentation efficiency, and attempted to achieve reasonable utilization of resources and sustainable economic development.

2. Material and methods

2.1. Experimental procedure and materials

Dried distiller's grain with solubles (DDGS) and sugarcane pith were obtained from Weifang Ensign Industry Co., Ltd (Shandong, China) and Nanxu sugar factory (Nanning, China), respectively. The physicochemical properties of fermentation raw materials are shown in Table 1. In order to obtain a suitable C/N ratio for solid state fermentation, DDGS was mixed with sugarcane pith at the ratio of 2:1, and the initial moisture was adjusted to about 60% (Sun et al., 2021b). An average of 1.75 kg of material was hand-packed into 5-L polyethylene terephthalate plastic canisters, producing a total of five plastic canisters. Four of the plastic tanks were sealed with three layers of polyethylene film, set as anaerobic fermentation (ANF). The remaining one was set as aerobic fermentation (AF). In this experiment, distillers dried grains with soluble were co-fermentation with carbon-rich sugarcane pith at an ambient temperature range of 18.5 ℃ to 30 ℃ for 50 d.

During the whole aerobic fermentation process, samples were fully mixed and homogeneous feed samples were collected by multi-point sampling method at 0, 4, 16, 25 and 50 days after fermentation initiation (López-González et al., 2015). Four anaerobic fermentation tanks were placed for 4, 16, 25 and 50 days, respectively, and the sampling method was the same as that of aerobic fermentation. The samples extracted from aerobic fermentation and anaerobic fermentation were denoted as AF and ANF, respectively. Samples were frozen (-20℃) until further analysis.

Table 1
The physicochemical properties of fermentation raw materials.

RM	Moisture (%)	pH	OM (%)	TKN (%)	C/N
bagasse pith	8.83 ± 0.12	7.08 ± 0.01	93.72 ± 0.04	0.41 ± 0.01	131.34 ± 3.20
DDGS	50.98 ± 0.22	3.98 ± 0.03	93.50 ± 0.13	4.68 ± 0.02	11.61 ± 0.05

Note: RM - raw materials; OM - organic matter; TKN - total Kjeldahl nitrogen; C/N - the ratio of organic carbon to total Kjeldahl nitrogen.

2.2. Analysis of physicochemical parameters

During the fermentation process, the temperature (T) and room temperature (RT) at three horizontal points in the center of the plastic tank were monitored at 12:00 noon every day. The samples were dried to constant weight in an oven at 80℃ to determine the moisture content, and then burned in a muffle furnace at 550 ℃ for 5 h to determine the content of organic matter (OM). The pH value of the sample extracts was determined by pH meter (PHS-25, China).

2.3. Extraction and analysis of DOM

5 g fermentation sample was placed in a conical flask, 50 mL deionized water was added, and shaken for 24 h. The mixture was centrifuged at 6,000 RPM and then filtered by a 0.45 μm membrane. (He et al., 2011; Li et al., 2017). The filtrate was freeze-dried and stored at 4℃ for further analysis. The characteristics of DOM in different fermentation stages were obtained by using total organic carbon analyzer (TOC-LCPH, Shimadzu, Japan) and Fourier transform infrared spectroscopy (IRTracer-100, Shimadzu, Japan). According to Noda's method, 2d correlation spectra were obtained by using 2Dshige, V.1.0.0.0 software. The synchronous correlation spectra are composed of autocorrelation peaks on the main diagonal and cross peaks on the non-diagonal, which can be used to obtain the degree of interference of the molecular structure and the correlation between functional groups. Overlapping peaks were distinguished by asynchronous correlation spectra, and the sequence of changes of functional groups was analyzed by combining two correlation spectra (Niu et al., 2021).

2.4. Degradation analysis of lignocellulose

Van Soest method was used to determine the content of lignocellulose in fermentation samples (Van Soest et al., 1991). The value of acid decontaminated fiber minus the value of acid decontaminated lignin was the content of cellulose. The content of hemicellulose was obtained by subtracting the value of neutral decontamination fiber from that of acidic decontamination fiber. In addition, the content of lignin was determined by the difference between ash and acid decontamination fiber.

2.5. Microbial sequencing

FastDNA® Spin Kit for Soil (MP Biomedicals, USA) was used to extract total DNA from fermentation samples, and purity and integrity of DNA were detected by NanoDrop2000 and 1% AGAR gel electrophoresis, respectively (Dineen et al., 2010). The DNA extract was stored at −20 ℃ for further analysis. The V3-V4 hypervariable region of 16S rRNA was amplified by bacterial universal primers 338F(ACTCCTACGGGAGGCAGCAG) and 806R (GGACTACHVGGGTWTCTAAT). Fungal ITS amplicon sequencing used the ITS1F (CTTGGTCATTTAGAGGAAGTAA) and ITS2R (GCTGCGTTCTTCATCGATGC) universal primers. Then, amplified products were sent to Majorbio Bio-Pharm Technology Co., Ltd. (Shanghai, China) for sequencing using Miseq PE300 platform (Illumina, San Diego, CA, USA).

2.6. Statistics analysis

All experiments were conducted in triplicates. The physicochemical parameters data were analyzed by SPSS Statistics 22.0 and ORIGIN 9.0. On the Majorbio cloud platform (www.MajorBio.com), the distance-based redundancy analysis method was used to analyze the relationship among environmental factors, microbial community structure and DOM characteristics.

3. Results and discussion

3.1. Analysis of physicochemical parameters of samples during fermentation

Fig. 1 showed the changes of environmental factors (temperature, moisture, pH, OM) that affect the efficiency of DDGS and sugarcane pith during solid state fermentation at different oxygen levels. Temperature was the main factor affecting the composition of microbial communities during fermentation (Guan et al., 2018). As shown in Fig. 1a, mesophilic microorganisms used biodegradable organic matter to grow and release heat in the process of AF, so the temperature increased rapidly from 26.0℃ (day 0) to 48.1℃ (day 3). Due to the degradation of organic matter, the temperature of the materials in the fermenter decreased gradually from 50.1 ℃ (day 5) to 27.85 ℃ (day 16). After 21 days of fermentation, the reactor temperature was basically maintained at the ambient temperature, indicating that the microbial activity decreased gradually. In the process of ANF, the temperature of the material in the plastic tank was basically maintained at the ambient temperature, and there was no obvious change.

During the fermentation process, the change of water content in the culture substrate would have an important effect on the growth and metabolism of microorganisms. Fig. 1b showed the change in moisture during the fermentation process. In AF, moisture increased gradually from day 0 (53.33%) to day 16 (59.18%), which was slightly higher than that of ANF. It is worth noting that moisture of AF continued to decrease from day 16 to day 50 (52.91%), while moisture of ANF on day 50 (59.76%) was higher than that of AF. During the whole fermentation process, whether AF or ANF, moisture was always within the range suitable for microbial growth (Raghavarao et al., 2003).

Fig. 1c illustrated the change of pH during fermentation, the tends of AF treatment increased first and then decreased, while the tends of ANF treatment decreased gradually. The initial pH would affect the activity of microorganisms, and the growth and metabolism of microorganisms would affect the pH as fermentation progresses. In AF process, the pH value increased rapidly with the mineralization of nitrogen-containing organic compounds (proteins and amino acids) and decomposition of organic acids under the action of microorganisms (Wang et al., 2016; Jiang et al., 2019). Due to the release of ammonia and organic decomposition into organic acids, the pH decreased gradually. In ANF process, the decrease of pH value was related to the production of organic acids. (Zhang et al.,2019).

In addition, the changes of organic matter reflected the growth and reproduction of microorganisms in the substrate. As shown in Fig. 1d, in AF process, due to the degradation of OM by microorganisms, the content of OM decreased from 93.46% (day 0) to 87.68% (day 50). However, under anaerobic conditions, the reduction of OM content in ANF was much lower than that in AF, because the decrease of pH inhibited the activity of some microorganisms and reduced the degradation of organic matter.

3.2. Characteristics of DOM

3.2.1. Analysis of DOM content

In the process of fermentation, DOM dissolved in the liquid film at the interface between gas and solid, which could be directly absorbed and utilized by microorganisms to drive microbial community succession (Fernandez-Bayo et al., 2018). The variation trend of DOM content during DDGS and sugarcane pith co-fermentation on different oxygen levels were shown in Fig. 2a. At the beginning of fermentation, the content of DOM decreased from 32.61 mg/g (day 0) to 23.87 mg/g (day 4) in AF process, owing to DOM can be directly consumed by microorganisms and converted into other metabolites. The content of DOM on day 16 was higher than that on day 4, because lignocellulose and other refractory organic matter could be converted into DOM. Then, the content of DOM decreased to 14.14 mg/g on day 50. Compared with AF,

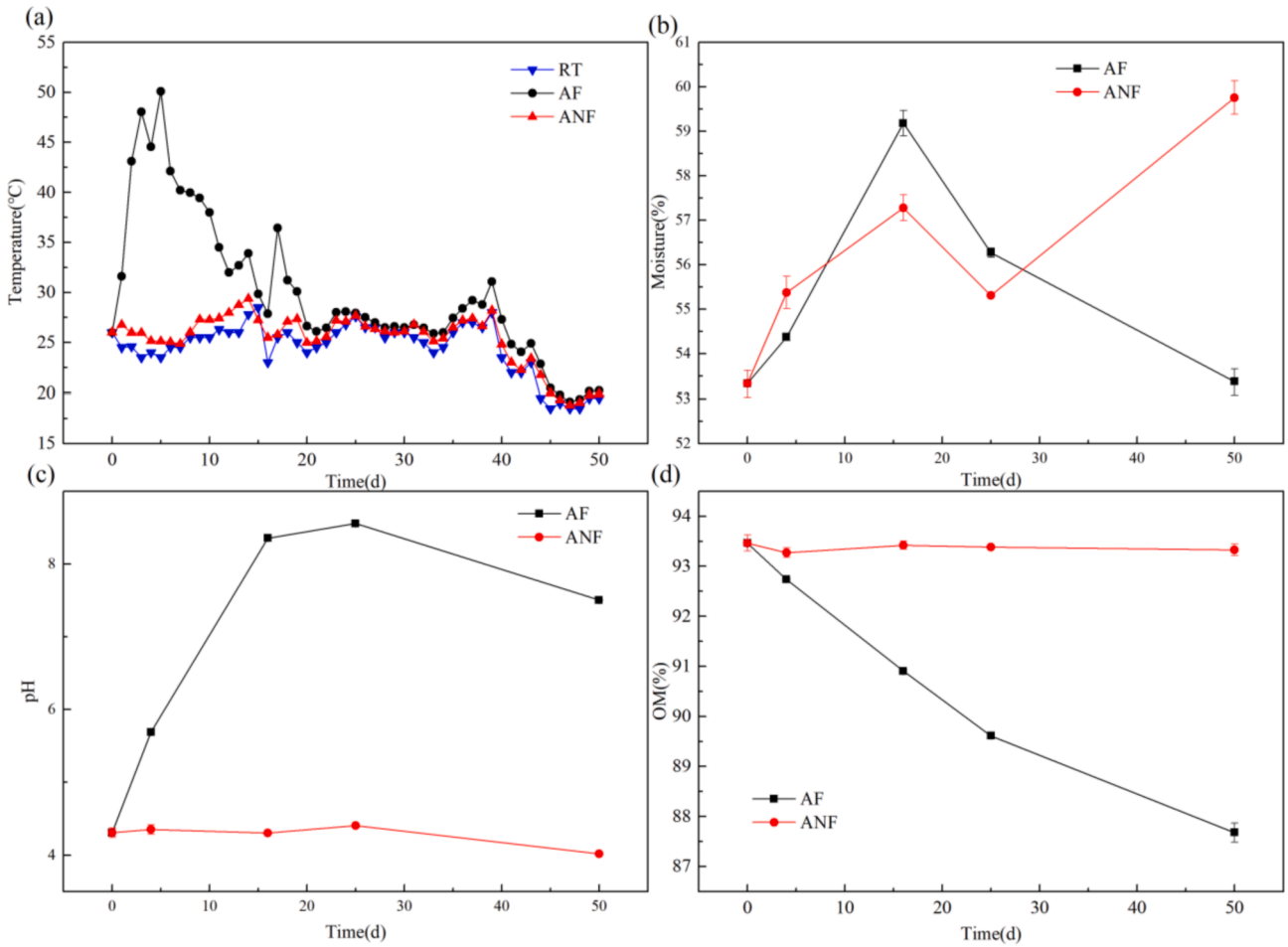


Fig. 1. Changes of temperature (a), Moisture (b), pH (c), OM (d) during fermentation process. Error bars indicate the standard deviation.

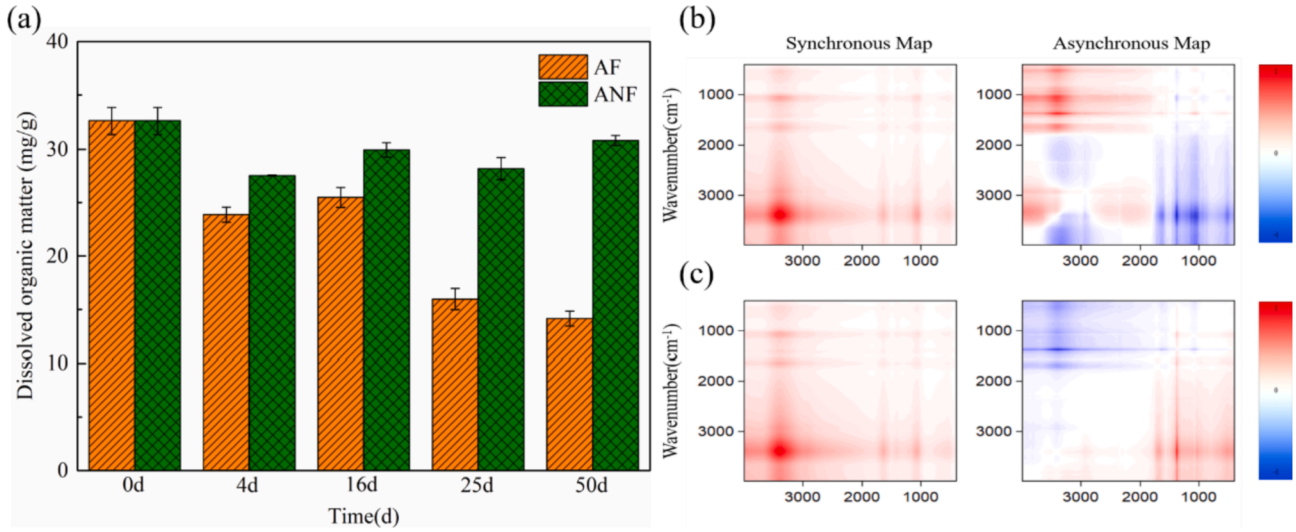


Fig. 2. Changes in the content of dissolved organic matter in the sample during fermentation (a). In the fermentation process of 0,4,16,25 and 50 days, synchronous and asynchronous maps with fermentation time as disturbance were generated from FTIR spectra of DOM: (b) AF in the 400–4000 cm^{-1} region;(c) ANF in 400–4000 cm^{-1} region. The colors range from blue to red, representing the correlation intensity from negative to positive. (For the reference materials on colors in this illustration, readers can refer to the online version of this article.) (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

the content of DOM decreases less in ANF. In conclusion, ANF was beneficial to the preservation of DOM content during co-fermentation process.

3.2.2. 2D-COS of FTIR analysis

Comparing the synchronous diagram of the co-fermentation process of DDGS and sugarcane pith at different oxygen levels (Fig. 2), the

transformation of DOM functional groups was more significant in Fig. 2b and weaker in Fig. 2c. Therefore, DOM changed more significantly in the presence of oxygen. All the cross peaks in the AF and ANF synchronization diagrams are positive, indicating that they were in the same direction of change. In detail, there were five main automatic peaks in the figure, located at 3400, 2923, 1645, 1407 and 1040 cm^{-1}, respectively, while the peak was at 3400 cm^{-1} was the most significant. The energy band near 3400 cm^{-1} may be due to O–H bonds. In addition, most automatic peaks on asynchronous maps have a plus or minus sign, indicating that DOM was also generated when DOM was consumed. Combined with the corresponding negative cross peaks in the AF asynchronous mapping, the structural change order of DOM functional groups was inferred: 3400 cm^{-1} > 2923 cm^{-1} > 1645 cm^{-1} > 1040 cm^{-1} > 1407 cm^{-1}, corresponding to O–H stretch vibration of hydroxyl groups(El Fels et al.,2014) > aliphatic C–H bond stretching vibration (Soobhany et al.,2017) > aromatic compound C = C bond stretching vibration (Zhu et al.,2020) > C-O bond stretching vibration of polysaccharides or polysaccharide–like substances (Zhang et al.,2014) > COO^- symmetric stretching and CH_3 bending vibrations: lipids, proteins (Aksoy and Severcan, 2012). Combined with the corresponding negative cross peaks in the ANF asynchronous mapping, the structural change order of DOM functional groups was inferred: 1407 cm^{-1} > 1040 cm^{-1} > 1645 cm^{-1} > 2923 cm^{-1} > 3400 cm^{-1}, which corresponds to COO^- symmetric stretching and CH_3 bending vibrations: lipids, proteins (Aksoy and Severcan, 2012) > C-O bond stretching vibration of polysaccharides or polysaccharide–like substances (Zhang et al.,2014) > stretching vibration of aromatic C = C bond (Zhu et al.,2020) > aliphatic C–H bond stretching vibration (Soobhany et al.,2017) > O–H stretch vibration of hydroxyl groups (El Fels et al.,2014).

To summarize, in the process of AF, phenols and alcohols in DOM were consumed first, followed by aliphatic, aromatic compounds, sugars and carbohydrates, and lipids and proteins. In the process of ANF, lipids and proteins in DOM were consumed first, followed by sugars and carbohydrates, aromatic compounds, aliphatic compounds, phenols and alcohols.

3.3. Degradation of lignocellulose

The special structure of lignin hindered the contact of microorganisms with cellulose and hemicellulose, so the degradation of lignocellulose became difficult (Wang et al., 2021). As shown in Fig. 3c, there was a significant difference in lignocellulose degradation between AF and ANF during co-fermentation of DDGS and sugarcane pith. Although at the initial stage of fermentation, the lignin degradation rate of AF was lower than that of ANF. However, with the extension of fermentation time, the lignin degradation rate of AF was significantly higher than that of ANF. After day 50, the lignin degradation rate of AF reached 36.50%. The accessibility of cellulase to lignocellulose increased significantly with the destruction of lignin structure (Wang et al., 2021). As a result, the degradation rates of cellulose and hemicellulose in AF were higher than those in ANF, which was consistent with the changes in Fig. 3a and 3b. In the process of microbial community succession, lignin was degraded into phenols and quinones, and cellulose was degraded into polysaccharides and monosaccharides, providing carbon and energy sources for microbial growth. Similarly, hemicellulose could also be absorbed and metabolized by microorganisms (Ma et al., 2020). The results showed that the microorganisms in the mixture of DDGS and sugarcane pith had stronger ability to degrade lignocellulose in AF,

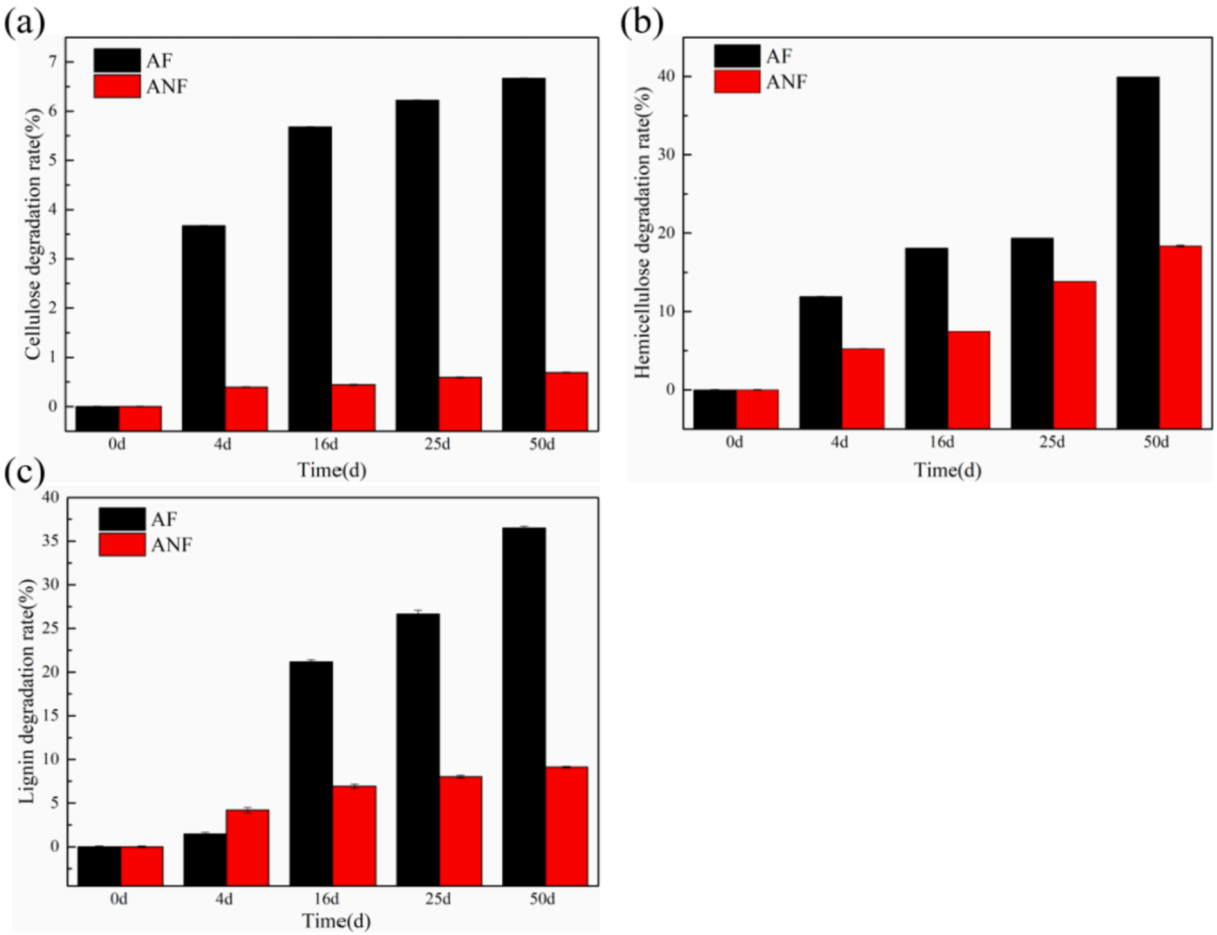


Fig. 3. The degradation rates of cellulose (a), hemicellulose (b) and lignin (c) during fermentation processes.

which was contrary to ANF. The degradation of lignocellulose led to the increase of DOM content, which was consistent with the change of DOM in 16 days. The purpose of this study was to compare the fermentation conditions of DDGS and sugarcane pith under different oxygen levels. According to the results, the fermentation with the endogenous microorganisms in the substrate has achieved certain effect, and the next step is to add some candidate strains fermentation (Sun et al., 2021b).

3.4. Microbial community analysis

3.4.1. Dynamics of bacterial community during co-fermentation

Fig. 4a showed the changes of phylum level bacterial communities during the co-fermentation of sugarcane pith and DDGS at different oxygen levels. At the initial stage, *Firmicutes, Actinobacteriota, Proteobacteria* and *Bacteroidota* were the dominant phyla in the fermentation process. During AF, the abundance of *Firmicutes* increased from day 0 (25.90%) to day 4 (98.32%) and decreased from day 4 (98.32%) to day 50 (8.64%). Liu et al., (2018) proved that *Firmicutes* can degrade lignocellulose and hydrolyze polysaccharides. Therefore, the change of *Firmicutes'* abundance resulted in the change of lignocellulose degradation rate during AF (Fig. 3). During ANF, the abundance of *Firmicutes* increased gradually from day 0 (25.90%) to day 50 (90.82%). Therefore, the degradation of lignocellulose in ANF process was relatively slow. During the AF process, *Actinobacteriota* decreased from 51.58% (day 0) to 0.71% (day 4), and then increased to 53.50% (day 50). *Actinobacteriotas* were mesophilic or thermophilic that participated in the degradation of lignocellulose residues, but only a small part of *Actinobacteriota* can tolerate high temperature during fermentation (Zhao et al., 2016; Zhang et al., 2020). In the ANF process, *Actinobacteriota* decreased from 51.58% (day 0) to 5.63% (day 50). The abundance of *Actinobacteriota* in AF was higher than that in ANF, resulting in a higher degradation rate of lignocellulose in AF. However, high microbial activity resulted in more DOM consumption in AF and reduced the nutritional value of fermentation products.

The changes of bacterial community at species level during co-fermentation of sugarcane pith and DDGS on different oxygen levels were shown in Fig. 4b. During AF, the abundance of *Bacillus ginsengihumi* increased from day 0 (3.32%) to day 4 (59.00%) and decreased from day 4 (59.00%) to day 50 (0.89%). The abundance of *Bacillus ginsengihumi* was the highest in AF on day 4, and that of *Bacillus ginsengihumi* was inhibited in ANF. *Bacillus ginsengihumi* could secrete phytase, which hydrolyzed phytic acid into inorganic phosphate and a small amount of phosphorylated inositol derivatives (Toymentseva et al., 2015; Troshagina et al., 2018). As a result, *Bacillus ginsengihumi* with high abundance was beneficial to the degradation of phytate phosphorus in DDGS and the digestion of DDGS feed by single-stomach animals, which improved the nutritional value of DDGS feed. In AF process, the abundance of *Lactobacillus plantarum* was always<0.15%, while in ANF process, the abundance of *Lactobacillus plantarum* increased gradually from day 0 (0.005%) to day 50(18.71%). Anaerobic environment and low pH were favorable for the growth of *Lactobacillus plantarum*. *Lactobacillus plantarum* had a high survival rate in acidic environment and could regulate intestinal flora and immune response of piglets. In addition, *Lactobacillus plantarum* could also secrete phytase, which had the ability to promote phytic acid degradation in DDGS (Dell'Anno et al., 2021; Shin et al., 2019; Sumengen et al., 2013; Śliżewska et al., 2021). In AF, the abundance of *Bacillus coagulans* increased from day 0 (0.38%) to day 4 (30.49%) and decreased from day 4 (30.49%) to day 50 (0.87%). The abundance of *Bacillus coagulans* was the highest on day 4, then decreased with the extension of fermentation time. In ANF, the abundance of *Bacillus coagulans* was always lower than 1.55%. *Bacillus coagulans* was a spore producing strain with high temperature resistance and characteristics of Bacillus and Lactobacillus, which has important application value in food industry. In addition, *Bacillus coagulans* improved growth performance of piglets by promoting intestinal barrier function and was a potential substitute for antibiotic growth promoters (Konuray and Erginkaya, 2018; Fu et al., 2021). Consequently, short-term AF fermentation was beneficial to increase the abundance of *Bacillus*

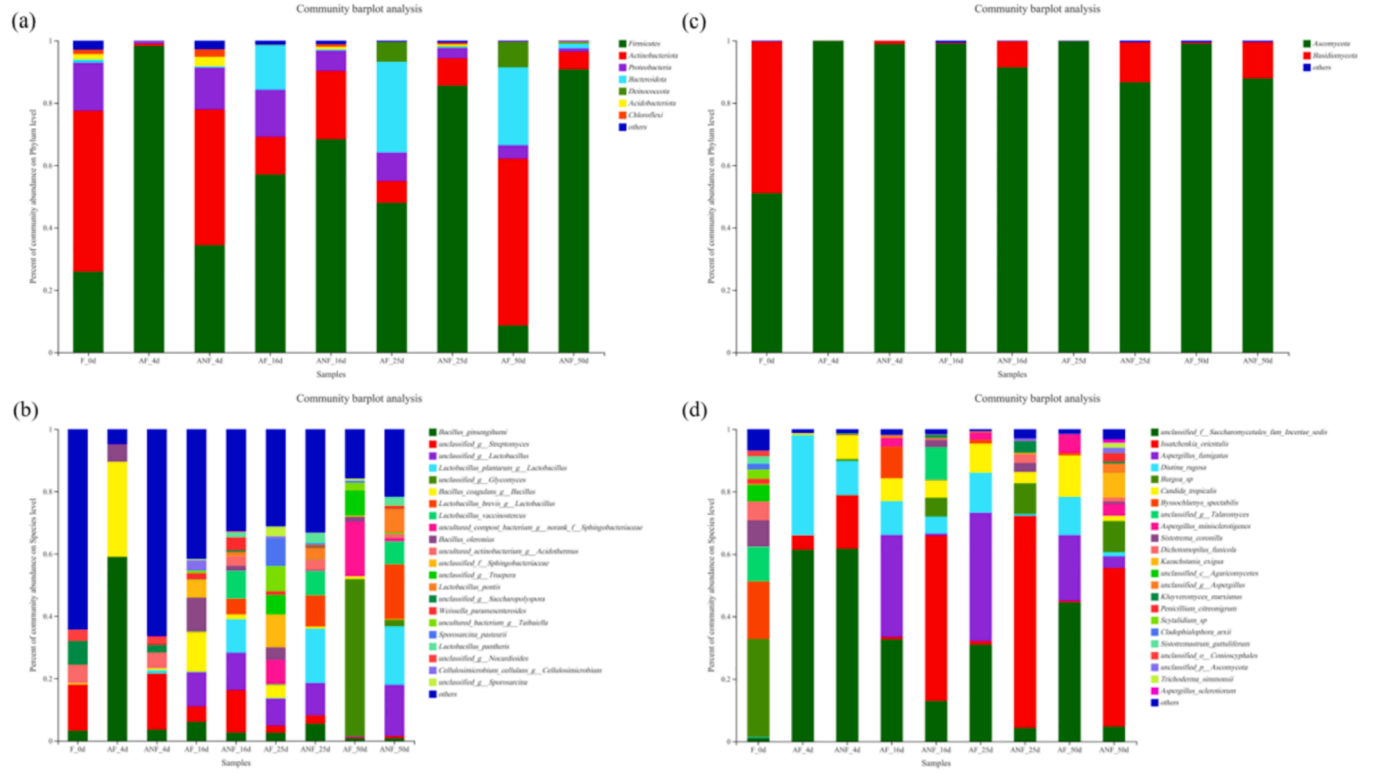


Fig. 4. Dynamic changes of microbial community during DDGS and sugarcane pith co-fermentation: The relative abundance of bacteria on phylum level (a) and species level (b); The relative abundance of fungi on phylum level (c) and species level (d).

coagulans and may improve the quality of feed. To sum up, short-term fermentation was beneficial to the growth of probiotics *Bacillus ginsengihumi* and *Bacillus coagulans* in AF (day 0-day 4), while long-term fermentation was beneficial to the growth of probiotics *Lactobacillus plantarum* in ANF (day 0-day 50).

3.4.2. Dynamics of fungal community during co-fermentation

The changes of phylum level fungal community during co-fermentation of sugarcane pith and DDGS at different oxygen levels were shown in Fig. 4c. At the beginning of fermentation, *Ascomycota* (51.08%) and *Basidiomycota* (48.69%) were the dominant phyla in the fungal community. In AF, *Ascomycota* was always the dominant phylum in the fermentation process, accounting for>99% of the total observed sequences. Different from AF, the abundance of *Ascomycota* in ANF gradually increased from day 0 (51.08%) to day 4 (98.96%) and decreased from day 4 (98.96%) to day 50 (87.91%). The abundance of *Basidiomycota* decreased gradually from day 0 (48.69%) to day 4 (1.03%), and increased slightly from day 4 (1.03%) to day 50 (11.69%). Therefore, *Ascomycota* was always the dominant fungal in the fermentation process under different oxygen levels.

The changes of fungal community at species level during co-fermentation of sugarcane pith and DDGS at different oxygen levels were shown in Fig. 4d. In the initial stage of fermentation, *Burgoa sp* (31.42%), *Byssochlamys spectabilis* (18.45%) and *unclassified Talaromyces* (11.11%) were the dominant bacteria in the fermentation samples. With the prolongation of fermentation time, the dominant species changed significantly during fermentation. In AF, the abundance of *Issatchenkia orientalis* increased from day 0 (0.00%) to day 4 (4.55%), then gradually decreased to day 50 (0.54%). In ANF, the abundance of *Issatchenkia orientalis* increased from day 0 (0.005%) to day 25 (67.82%), and then gradually decreased to day 50 (50.75%). *Issatchenkia orientalis* was a kind of non-yeast with multiple tolerance (for example, acetic acid tolerance, high heat, and salt tolerance) and had potential in wine making and bioethanol production (Li et al., 2020a, Li et al., 2020b). Therefore, in ANF, an increase in the abundance of *Issatchenkia orientalis* may produce a large amount of ethanol. In AF, the abundance of *Aspergillus fumigatus* was low in the first four days (≤0.04%), but with the extension of fermentation time, the abundance of *Aspergillus fumigatus* increased, which was not conducive to the health of animals. In ANF, *Aspergillus fumigatus* abundance was low, with day 50 (3.75%) reaching its maximum. The results showed that the control of fermentation time was beneficial to reduce the pollution of feed products. In AF, the abundance of *Diutina rugosa* increased from day 0 (0.22%) to day 4 (32.09%) and then decreased to day 50 (12.21%). In ANF, the abundance of *Diutina rugosa* increased from day 0 (0.22%) to day 4 (10.96%), then gradually decreased to day 50 (1.31%). *Diutina rugosa* can be used as a feed additive to regulate immunity and promote the growth of broilers (Wang et al., 2019). In AF, the abundance of *Candida tropicalis* gradually increased from day 0 (0.003%) to day 50 (13.28%), while in ANF, the abundance of *Candida tropicalis* gradually increased from day 0 (0.003%) to day 4 (7.63%) and decreased from day 4 (7.63%) to day 50 (1.65%). *Candida tropicalis* had the ability to produce extracellular phytase (Suryani et al., 2021). Therefore, *Candida Tropicalis* was beneficial to the degradation of phytic acid in DDGS.

3.5. Distance-based redundancy analysis

As shown in Fig. 5, the relationship between environmental factors (temperature, moisture, pH, OM, DOM) and microbial communities (bacteria and fungi) at species level was investigated by distance-based redundancy analysis. The factor that has the great influence on bacterial community and species distribution was pH ($R = 0.8505$, $P < 0.01$), followed by OM ($R = 0.6576$, $P < 0.05$), DOM ($R = 0.6465$, $P < 0.05$), moisture and temperature, as shown in Fig. 5a. Specifically, DOM was positively correlated with OM, but negatively correlated with pH, moisture and temperature, indicating that DOM was consumed by microorganisms with the degradation of organic matter, and the increase of temperature, pH and moisture resulted in the decrease of DOM. In addition, the relationship between bacterial species and environmental factors in the first five abundances was analyzed, and it was considered that environmental factors had great effects on *Bacillus ginsengihumi* and *Lactobacillus plantarum*. *Bacillus ginsengihumi* was positively correlated with temperature and pH, indicating that the increase of temperature and pH value in the middle and early stage of AF was beneficial to the activity of *Bacillus ginsengihumi*. *Lactobacillus plantarum* was positively correlated with moisture, indicating that the increase of moisture in ANF was beneficial to the growth of *Lactobacillus plantarum*. In conclusion, organic matter produced soluble organic matter under the metabolism of bacterial community, but moisture, temperature and pH value affected the abundance of bacterial community and indirectly affected the characteristics of DOM. In addition, the growth and metabolism of microorganisms also consumed part of DOM, which resulted in the diversity of DOM content and functional groups.

In order to explore the relationship between microorganisms and environmental factors in more detail, fungal communities were also discussed. As shown in Fig. 5b, the factor that has great influence on fungal community and species distribution was pH ($R = 0.6308$, $P < 0.05$), followed by DOM ($R = 0.6223$, $P < 0.05$), OM, moisture and temperature. In detail, DOM was positively correlated with OM, moisture, and negatively correlated with pH and temperature. Interestingly,

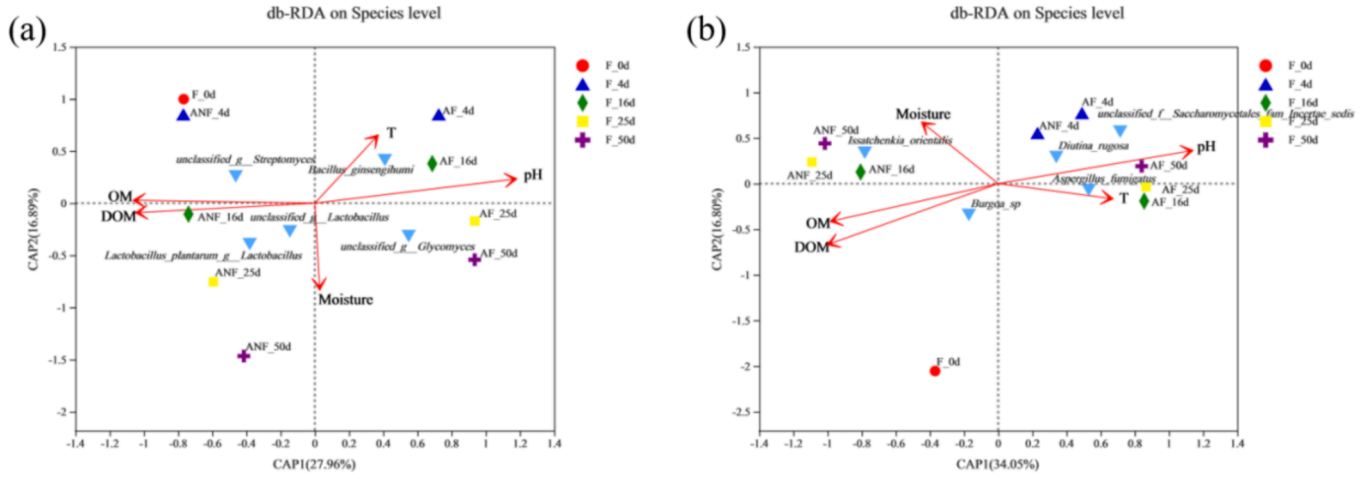


Fig. 5. Distance-based redundancy analysis of: environment factors and bacterial community (a); environment factors and fungi community (b). T: temperature. OM: organic matter. DOM: dissolved organic matter.

it was slightly different from the correlation of the above bacterial communities. The relationship between the top five fungi species and environmental factors was analyzed, which indicated that environmental factors had a great influence on *Issatchenkia orientalis*. *Issatchenkia orientalis* was positively correlated with water content, OM and DOM, and negatively correlated with pH and temperature, indicating that *Issatchenkia orientalis* promoted the generation of DOM. However, the increase of pH and temperature would inhibit the abundance of *Issatchenkia orientalis* and control the generation of DOM.

4. Conclusions

This study revealed that ANF was beneficial to the preservation of DOM content during fermentation process. Phenols and alcohols were consumed first in AF, while lipids and proteins were consumed first in ANF. *Bacillus ginsengihumi* and *Issatchenkia orientalis* played an important role in the degradation of OM and the formation of DOM. In addition, the main environmental factors driving the microbial community were pH, OM and DOM in the process of fermentation. In summary, these findings provided some novel insights into the signature of dissolved organic matter and relevant microbiome during the co-fermentation of DDGS and sugarcane pith.

CRediT authorship contribution statement

Yiwu Wang: Conceptualization, Methodology, Investigation, Writing – original draft. **Xiaolan Li:** Data curation, Investigation. **Kecheng Li:** Data curation, Investigation, Writing – review & editing. **Yite Huang:** Data curation, Investigation. **Hongxiang Yang:** Data curation, Investigation. **Pengfei Zhu:** Data curation, Investigation. **Zhanyou Chi:** Writing – review & editing. **Yongping Xu:** Writing – review & editing. **Qunliang Li:** Conceptualization, Supervision, Funding acquisition, Writing – review & editing.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

The authors thank other members of our laboratory for their constructive advice and useful discussions, as well as the funding of the National Natural Science Foundation of China (No. 41861124004, No. 21878057).

References

Aksoy, C., Severcan, F., 2012. Role of vibrational spectroscopy in stem cell research. J. Spectroscopy. 27 (3), 167–184. https://doi.org/10.1155/2012/513286.

Chen, W., Teng, C.Y., Qian, C., Yu, H.Q., 2019. Characterizing Properties and Environmental Behaviors of Dissolved Organic Matter Using Two-Dimensional Correlation Spectroscopic Analysis. Environ. Sci. Technol. 53 (9), 4683–4694. https://doi.org/10.1021/acs.est.9b01103.

Dell'Anno, M., Callegari, M.L., Reggi, S., Caprarulo, V., Giromini, C., Spalletta, A., Coranelli, S., Sgoifo Rossi, C.A., Rossi, L., 2021. Lactobacillus plantarum and Lactobacillus reuteri as Functional Feed Additives to Prevent Diarrhoea in Weaned Piglets. Animals (Basel). 11 (6), 1766. https://doi.org/10.3390/ani11061766.

Dineen, S.M., Aranda, R., Anders, D.L., Robertson, J.M., 2010. An evaluation of commercial DNA extraction kits for the isolation of bacterial spore DNA from soil. J. Appl. Microbiol. 109 (6), 1886–1896. https://doi.org/10.1111/j.1365-2672.2010.04816.x.

El Fels, L., Zamama, M., El Asli, A., Hafidi, M., 2014. Assessment of biotransformation of organic matter during co-composting of sewage sludge-lignocelullosic waste by chemical, FTIR analyses, and phytotoxicity tests. Int. Biodeter. Biodegr. 87, 128–137. https://doi.org/10.1016/j.ibiod.2013.09.024.

Fasching, C., Akotoye, C., Bižić, M., Fonvielle, J., Ionescu, D., Mathavarajah, S., Zoccarato, L., Walsh, D.A., Grossart, H.P., Xenopoulos, M.A., 2020. Linking stream microbial community functional genes to dissolved organic matter and inorganic nutrients. Limnol. Oceanogr. 65, S71–S87. https://doi.org/10.1002/lno.11356.

Fernandez-Bayo, J.D., Yazdani, R., Simmons, C.W., VanderGheynst, J.S., 2018. Comparison of thermophilic anaerobic and aerobic treatment processes for stabilization of green and food wastes and production of soil amendments. Waste. Manag. 77, 555–564. https://doi.org/10.1016/j.wasman.

Fu, R., Liang, C., Chen, D., Yan, H., Tian, G., Zheng, P., He, J., Yu, J., Mao, X., Huang, Z., Luo, Y., Luo, J., Yu, B., 2021. Effects of dietary Bacillus coagulans and yeast hydrolysate supplementation on growth performance, immune response and intestinal barrier function in weaned piglets. J. Anim. Physiol. Anim. Nutr (Berl) 105 (5), 898–907. https://doi.org/10.1111/jpn.13529.

Guan, H., Yan, Y., Li, X., Li, X., Shuai, Y., Feng, G., Ran, Q., Cai, Y., Li, Y., Zhang, X., 2018. Microbial communities and natural fermentation of corn silages prepared with farm bunker-silo in Southwest China. Bioresour. Technol. 265, 282–290. https://doi.org/10.1016/j.biortech.2018.06.018.

He, X.S., Xi, B.D., Wei, Z.M., Jiang, Y.H., Geng, C.M., Yang, Y., Yuan, Y., Liu, H.L., 2011. Physicochemical and spectroscopic characteristics of dissolved organic matter extracted from municipal solid waste (MSW) and their influence on the landfill biological stability. Bioresour. Technol. 102 (3), 2322–2327. https://doi.org/10.1016/j.biortech.2010.10.085.

Iram, A., Cekmecelioglu, D., Demirci, A., 2020. Distillers' dried grains with solubles (DDGS) and its potential as fermentation feedstock. Appl. Microbiol. Biotechnol. 104 (14), 6115–6128. https://doi.org/10.1007/s00253-020-10682-0.

Iram, A., Cekmecelioglu, D., Demirci, A., 2021. Screening of bacterial and fungal strains for cellulase and xylanase production using distillers' dried grains with solubles (DDGS) as the main feedstock. Biomass. Conv. Bioref. 11 (5), 1955–1964. https://doi.org/10.1007/s13399-019-00588-x.

Jiang, Z., Lu, Y., Xu, J., Li, M., Shan, G., Li, Q., 2019. Exploring the characteristics of dissolved organic matter and succession of bacterial community during composting. Bioresour. Technol. 292, 121942. https://doi.org/10.1016/j.biortech.2019.121942.

Konuray, G., Erginkaya, Z., 2018. Potential Use of Bacillus coagulans in the Food Industry. Foods. 7 (6), 92. https://doi.org/10.3390/foods7060092.

Li, H.Y., Wang, H., Wang, H.T., Xin, P.Y., Xu, X.H., Ma, Y., Liu, W.P., Teng, C.Y., Jiang, C. L., Lou, L.P., Arnold, W., Cralle, L., Zhu, Y.G., Chu, J.F., Gilbert, J.A., Zhang, Z.J., 2018. The chemodiversity of paddy soil dissolved organic matter correlates with microbial community at continental scales. Microbiome. 6 (1), 187. https://doi.org/10.1186/s40168-018-0561-x.

Li, S., Jin, Z., Hu, D., Yang, W., Yan, Y., Nie, X., Lin, J., Zhang, Q., Gai, D.i., Ji, Y., Chen, X., 2020a. Effect of solid-state fermentation with Lactobacillus casei on the nutritional value, isoflavones, phenolic acids and antioxidant activity of whole soybean flour. LWT. 125, 109264. https://doi.org/10.1016/j.lwt.2020.109264.

Li, Q., Lu, Y., Guo, X., Shan, G., Huang, J., 2017. Properties and evolution of dissolved organic matter during co-composting of dairy manure and Chinese herbal residues. Environ. Sci. Pollut. Res. Int. 24 (9), 8629–8636. https://doi.org/10.1007/s11356-017-8540-5.

Li, Y., Wu, Z., Li, R., Miao, Y., Weng, P., Wang, L., 2020b. Integrated transcriptomic and proteomic analysis of the acetic acid stress in Issatchenkia orientalis. J. Food. Biochem. 44 (6), e13203 https://doi.org/10.1111/jfbc.13203.

Liu, L., Wang, S., Guo, X., Zhao, T., Zhang, B., 2018. Succession and diversity of microorganisms and their association with physicochemical properties during green waste thermophilic composting. Waste. Manag. 73, 101–112. https://doi.org/10.1016/j.wasman.2017.12.026.

López-González, J.A., Suárez-Estrella, F., Vargas-García, M.C., López, M.J., Jurado, M. M., Moreno, J., 2015. Dynamics of bacterial microbiota during lignocellulosic waste composting: Studies upon its structure, functionality and biodiversity. Bioresour. Technol. 175, 406–416. https://doi.org/10.1016/j.biortech.2014.10.123.

Ma, C., Lo, P.K., Xu, J., Li, M., Jiang, Z., Li, G., Zhu, Q., Li, X., Leong, S.Y., Li, Q., 2020. Molecular mechanisms underlying lignocellulose degradation and antibiotic resistance genes removal revealed via metagenomics analysis during different agricultural wastes composting. Bioresour. Technol. 314, 123731. https://doi.org/10.1016/j.biortech.2020.123731.

Muhammad, A.K., Muhammad, Z.K., Khalid, Z., Lubna, N., 2014. Global estimates of energy consumption and greenhouse gas emissions. Renew. Sust. Energ. Rev. 29, 336–344. https://doi.org/10.1016/j.rser.2013.08.091.

Nebbioso, A., Piccolo, A., 2013. Molecular characterization of dissolved organic matter (DOM): a critical review. Anal. Bioanal. Chem. 405 (1), 109–124. https://doi.org/10.1007/s00216-012-6363-2.

Niu, Q., Meng, Q., Yang, H., Wang, Y., Li, X., Li, G., Li, Q., 2021. Humification process and mechanisms investigated by Fenton-like reaction and laccase functional expression during composting. Bioresour. Technol. 341, 125906. https://doi.org/10.1016/j.biortech.2021.125906.

Raghavarao, K.S.M.S., Ranganathan, T.V., Karanth, N.G., 2003. Some engineering aspects of solid-state fermentation. Biochem. Eng. J. 13 (2-3), 127–135. https://doi.org/10.1016/S1369-703X(02)00125-0.

Shin, D., Chang, S.Y., Bogere, P., Won, KyeongHye, Choi, J.-Y., Choi, Y.-J., Lee, H.K., Hur, J., Park, B.-Y., Kim, Y., Heo, J., Tassou, C., 2019. Beneficial roles of probiotics on the modulation of gut microbiota and immune response in pigs. PLoS ONE 14 (8), e0220843. https://doi.org/10.1371/journal.pone.0220843.

Śliżewska, K., Chlebicz-Wójcik, A., Nowak, A., 2021. Probiotic Properties of New Lactobacillus Strains Intended to Be Used as Feed Additives for Monogastric Animals. Probiotics. Antimicrob. Proteins. 13 (1), 146–162. https://doi.org/10.1007/s12602-020-09674-3.

Soobhany, N., Gunasee, S., Rago, Y.P., Joyram, H., Raghoo, P., Mohee, R., Garg, V.K., 2017. Spectroscopic, thermogravimetric and structural characterization analyses for comparing Municipal Solid Waste composts and vermicomposts stability and maturity. Bioresour. Technol. 236, 11–19. https://doi.org/10.1016/j.biortech.2017.03.161.

Sumengen, M., Dincer, S., Kaya, A., 2013. Production and characterization of phytase from Lactobacillus plantarum. Food. Biotechnol. 27 (2), 105–118. https://doi.org/10.1080/08905436.2013.781507.

Sun, X., Chen, Y., Luo, L., Heidari, F., Tiffany, D.G., Urriola, P.E., Shurson, G.G., Bo, H.u., 2021a. Feeding value improvement by co-fermentation of corn-ethanol co-product and agro-industrial residues with Rhizopus oryzae. Process. Biochem. 111 (2), 144–153. https://doi.org/10.1016/j.procbio.2021.10.029.

Sun, X., Tiffany, D.G., Urriola, P.E., Shurson, G.G., Hu, B.o., 2021b. Nutrition upgrading of corn-ethanol co-product by fungal fermentation: Amino acids enrichment and anti-nutritional factors degradation. Food. Bioprod. Process. 130, 1–13. https://doi.org/10.1016/j.fbp.2021.09.004.

Suominen, S., Dombrowski, N., Sinninghe Damsté, J.S., Villanueva, L., 2021. A diverse uncultivated microbial community is responsible for organic matter degradation in the Black Sea sulphidic zone. Environ. Microbiol. 23 (6), 2709–2728. https://doi.org/10.1111/1462-2920.14902.

Suryani, A.E., Anggraeni, A.S., Istiqomah, L., Damayanti, E., Karimy, M.F., 2021. Isolation and identification of phytate-degrading yeast from traditional fermented food. Biodiversitas. 22 (2). https://doi.org/10.13057/biodiv/d220241.

Toymentseva, A.A., Suleimanova, A.D., Boulygina, E.A., Kazakov, S.V., Baranova, D.S., Akhmetova, A.I., Mardanova, A.M., Sharipova, M.R., 2015. Draft Genome Sequence of Bacillus ginsengihumi Strain M2.11 with Phytase Activity. Genome. Announc. 3 (4), e00851–e915. https://doi.org/10.1128/genomeA.00851-15.

Troshagina, D.S., Suleimanova, A.D., Itkina, D.L., Sharipova, M.R., 2018. Cloning of Phytase Genes from Pantoea Sp. 3.5.1 and Bacillus ginsengihumi M2.11 in Pichia pastoris. BioNanoSci. 8 (4), 1045–1053.

Van Soest, P.J., Robertson, J.B., Lewis, B.A., 1991. Methods for dietary fiber, neutral detergent fiber, and nonstarch polysaccharides in relation to animal nutrition. J. Dairy. Sci. 74 (10), 3583–3597. https://doi.org/10.3168/jds.S0022-0302(91)78551-2.

Wang, C., Guo, X., Deng, H., Dong, D., Tu, Q., Wu, W., 2014. New insights into the structure and dynamics of actinomycetal community during manure composting. Appl. Microbiol. Biotechnol. 98 (7), 3327–3337. https://doi.org/10.1007/s00253-013-5424-6.

Wang, J., Wang, B., Du, H., Zhang, H., Li, H., Wang, F., Zhao, X., 2019. Effects of Diutina rugosa SD-17 on growth performance, intestine morphology, and immune status of chickens. Poult. Sci. 98 (12), 6311–6318. https://doi.org/10.3382/ps/pez428.

Wang, Q., Wang, Z., Awasthi, M.K., Jiang, Y., Li, R., Ren, X., Zhao, J., Shen, F., Wang, M., Zhang, Z., 2016. Evaluation of medical stone amendment for the reduction of nitrogen loss and bioavailability of heavy metals during pig manure composting. Bioresour. Technol. 220, 297–304. https://doi.org/10.1016/j.biortech.2016.08.081.

Wang, S., Meng, Q., Zhu, Q., Niu, Q., Yan, H., Li, K., Li, G., Li, X., Liu, H., Liu, Y., Li, Q., 2021. Efficient decomposition of lignocellulose and improved composting performances driven by thermally activated persulfate based on metagenomics analysis. Sci. Total. Environ. 794, 148530. https://doi.org/10.1016/j.scitotenv.2021.148530.

Wu, B.o., Wang, Y.-W., Dai, Y.-H., Song, C., Zhu, Q.-L., Qin, H., Tan, F.-R., Chen, H.-C., Dai, L.-C., Hu, G.-Q., He, M.-X., 2021. Current status and future prospective of bio-ethanol industry in China. Renew. Sust. Energ. Rev. 145, 111079. https://doi.org/10.1016/j.rser.2021.111079.

Yang, L., Zeng, X., Qiao, S., 2021. Advances in research on solid-state fermented feed and its utilization: The pioneer of private customization for intestinal microorganisms. Anim. Nutr. 7 (4), 905–916. https://doi.org/10.1016/j.aninu.2021.06.002.

Zabed, H., Sahu, J.N., Suely, A., Boyce, A.N., Faruq, G., 2017. Bioethanol production from renewable sources: Current perspectives and technological progress. Renew. Sust. Energ. Rev. 71, 475–501. https://doi.org/10.1016/j.rser.2016.12.076.

Zhao, Y., Lu, Q., Wei, Y., Cui, H., Zhang, X., Wang, X., Shan, S., Wei, Z., 2016. Effect of actinobacteria agent inoculation methods on cellulose degradation during composting based on redundancy analysis. Bioresour. Technol. 219, 196–203. https://doi.org/10.1016/j.biortech.2016.07.117.

Zhang, C., Gao, Z., Shi, W., Li, L., Tian, R., Huang, J., Lin, R., Wang, B., Zhou, B.o., 2020. Material conversion, microbial community composition and metabolic functional succession during green soybean hull composting. Bioresour. Technol. 316, 123823. https://doi.org/10.1016/j.biortech.2020.123823.

Zhang, J., Lü, F., Shao, L., He, P., 2014. The use of biochar-amended composting to improve the humification and degradation of sewage sludge. Bioresour. Technol. 168, 252–258. https://doi.org/10.1016/j.biortech.2014.02.080.

Zhang, L.u., Zhou, X., Gu, Q., Liang, M., Mu, S., Zhou, B.o., Huang, F., Lin, B.o., Zou, C., 2019. Analysis of the correlation between bacteria and fungi in sugarcane tops silage prior to and after aerobic exposure. Bioresour. Technol. 291, 121835. https://doi.org/10.1016/j.biortech.2019.121835.

Zhu, Q., Li, G., Jiang, Z., Li, M., Ma, C., Li, X., Li, Q., 2020. Investigating the variation of dissolved organic matters and the evolution of autotrophic microbial community in composting with organic and inorganic carbon sources. Bioresour. Technol. 304, 123013. https://doi.org/10.1016/j.biortech.2020.123013.